

HEATMAP NODE · REQUIREMENTS

HeatMap Node requirements

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DOCUMENT CONTROL

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Project TRL	TRL 2 Technology concept formulated		
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REVISION HISTORY

Version	Date	Author	Change
0.2	2026-09-25	Amish Chadha	First measurable requirements for TRL 2, with status against the concept estimates
0.1	2026-09-25	Amish Chadha	Initial scaffold

HeatMap Node requirements

These are first-pass requirements for the concept. Targets are proposals for review, not yet validated with users, and will be checked by calculation at TRL 3 and revised after co-design sessions (see HMN-PRB-001). Status is judged against the first-order estimates in HMN-PRC-001; nothing has been tested.

Three requirements are **not met** by the concept as drawn (R4 at low wind, R9 and R13 for the full node), and three are **at risk** (R1, R5 and R15). See Table 2.

Table 1. Requirements.

ID	Requirement	Target	Verification (TRL 3 or later)
R1	Air temperature accuracy in the street	± 0.5 °C against a reference aspirated thermometer, in full sun at wind speeds of 1 m/s or more; ± 1.0 °C below 1 m/s	Shield error estimate; later side-by-side field comparison
R2	Relative humidity accuracy	± 3 %RH from 20 to 80 %RH at 15 to 40 °C	Datasheet; CalRig salt fixed points
R3	Globe temperature	Standard 150 mm black globe (emissivity about 0.95); probe ± 0.3 °C after calibration; 90 % response within 30 min	Calibration in CalRig; response estimate
R4	Wind speed	0.5 to 20 m/s; ± 0.5 m/s or ± 10 %, whichever is larger	Datasheet of the chosen anemometer; later wind comparison
R5	Derived heat stress	Mean radiant temperature and outdoor WBGT computed per ISO 7243 definitions with the Liljegren method; WBGT within ± 1.0 °C of a reference WBGT meter	Calculation note; later collocation with a reference meter
R6	Reporting	15 min means of all channels by default (adjustable 5 to 60 min); data at the gateway within 30 min; 7 days stored on the node for resend	FieldNode firmware design; airtime calculation
R7	Power autonomy	No mains; 5 days without sun at the default reporting interval	FieldNode energy budget plus the sensor load in HMN-PRC-001
R8	Outdoor survival	Electronics IP65; operate at -20 to +60 °C at the node; UV-stable plastics; sensors survive driving rain and dust	Design review; datasheets
R9	Measurement height	Sensors at pedestrian height (1.1 to 2.0 m)	Mounting review with pole owners
R10	Fit to existing poles	Stainless band clamps for 60 to 200 mm round poles; no drilling; fitted from a ladder or lift by two people in 30 min or less	Design review; later timed trial
R11	Wind survival	No failure at a 35 m/s gust; arm deflection under 10 mm at 20 m/s	Wind load calculation
R12	Privacy	Environmental data only: no camera, microphone, Wi-Fi or Bluetooth scanning, or any personal data	Design review of hardware and firmware
R13	Cost	Sensor head \$120 or less; full node including the FieldNode core \$120 or less (the <code>budget_usd</code> in <code>project.yaml</code>)	Priced BOM
R14	Traceable data	Each node carries a calibration record from CalRig; readings published in an open, documented format with node location and height	Documentation review
R15	Mass and loading on the pole	4 kg or less for the complete node; no part larger than 300 mm across other than the arm	Mass estimate

Table 2. Status against the concept estimates (HMN-PRC-001).

ID	Estimate	Status
R1	Passive multi-plate shields can read about 1 °C or more high in strong sun and calm air (estimate, to be quantified)	At risk below about 1 m/s
R2	SHT45-class sensor behind a membrane	Met by design, to be confirmed
R3	150 mm globe, NTC at center; response time about 20 to 30 min (estimate)	Met by design, to be confirmed
R4	Low-cost cup anemometers typically start at about 0.8 m/s (to be confirmed)	Not met from 0.5 to 0.8 m/s
R5	A 0.5 m/s wind error at 1 m/s shifts mean radiant temperature by about 6 to 8 °C in the worked example	At risk in calm air
R6	96 uplinks a day at about 20 B	Met by design
R7	Sensor load under 1 mW against about 115 mW allowed by FieldNode	Met, with a wide margin
R8	Stock IP65 enclosure and M12 connectors	Met by design, to be confirmed
R9	Arm at about 2.8 m to limit tampering	Not met ; correction or a lower arm proposed, awaiting Amish
R10	Longer bands than FieldNode's 40 to 60 mm kit	Met by design, to be confirmed
R11	About 110 N total at 35 m/s; about 17 N·m at the arm root	Met by estimate
R12	No imaging or audio parts in the BOM	Met
R13	Sensor head about \$120; full node about \$246	Sensor head met with no margin; full node not met
R14	CalRig procedure not yet written	Open
R15	About 3.3 kg	Met; at risk if a larger panel is needed in cloudy climates

■ Assumptions

- The FieldNode core performs as stated in its README and precis (about 115 mW sensor allowance, 5 days without sun).
- WBGT is derived, not measured with a wetted wick, because a natural wet-bulb sensor needs a water reservoir that would need refilling at every node.
- Pole owners permit clamped attachments at the proposed height.